

Research Methods — Module 07: Communication & Synthesis — Study Questions

Part 1: Fundamental Comprehension

1. **The Interdisciplinary Core:** Why is Active Inference considered an inherently interdisciplinary framework? Which specific epistemic gaps does it attempt to bridge between physics, biology, and cognitive science?
2. **Rosetta Stone:** What constitutes the "shared formal language" (Friston, 2010) that allows researchers from disparate disciplines to collaborate on Active Inference projects? Explain this mathematical ontology.
3. **The Unfalsifiability Critique:** Explain the common critique that the Free Energy Principle is unfalsifiable. How do proponents of the framework differentiate between the foundational *principle* (a tautology of survival) and specific falsifiable *process theories*?
4. **Bayesian Brain Clarifications:** What are the critical nuances that popular science often misses when claiming "the brain literally calculates Bayes' theorem"? Distinguish between exact Bayesian inference and variational approximations.
5. **Open Science Infrastructure:** How do open-source computational ecosystems (e.g., `pymdp`, `SPM`, `RxInfer.jl`) accelerate transdisciplinary Active Inference research? What specific role does each package play in the research lifecycle?

Part 2: Advanced Analysis & Synthesis

1. **Epistemic Clashes:** What specific ontological or methodological challenges typically erupt when cellular biologists collaborate with theoretical physicists on an Active Inference model of a cell? How are these resolved mathematically?
2. **Jargon Translation:** Translate the concept of "precision-weighted prediction error" into language appropriate for: (a) a practicing clinical psychiatrist, (b) a machine learning engineer building reinforcement learning agents, and (c) a layperson reading a popular science article.

3. **The Role of Philosophy:** What is the structural role of philosophy of mind (specifically enactivism and phenomenology) in driving empirical Active Inference research? Give an example of a philosophical constraint informing a mathematical model.
4. **Meta-Science:** How does the structure of the Active Inference Institute itself reflect the principles of distributed intelligence and stigmergic communication discussed within the theory?
5. **The Literate Programming Paradigm:** Argue for or against the necessity of "literate programming" (intertwining prose, mathematics, and executable code in a single document like a Jupyter Notebook) as a requirement for publishing reproducible Active Inference simulations.

Part 3: Methodological Application

1. **Preregistration Architecture:** Draft a brief preregistration plan for a computational modeling study comparing Active Inference agents to classic Q-learning agents in a volatile reward environment. What parameters must be locked in advance?
2. **Curriculum Design:** Design a highly compressed, 2-day workshop syllabus for teaching the required mathematical prerequisites (linear algebra, probability, information theory) to psychology graduate students entering an Active Inference lab.
3. **Data Sharing Specifications:** If you were establishing a multi-site consortium for computational psychiatry using Active Inference, what specific behavioral and neuroimaging data formats would you enforce to ensure seamless ingestion into generative models?
4. **Peer Review Simulation:** You are reviewing a paper that claims to "prove the Free Energy Principle using fMRI." Based on your understanding of the methodology, what are the primary fatal flaws you expect to find in their experimental design or theoretical framing?
5. **Cross-Disciplinary Grant Writing:** Outline a grant proposal attempting to secure funding for a team consisting of an ethnographer, a neuroscientist, and a computational modeler, using Active Inference as the unifying methodology to study collective rituals.